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TRANSPORT PHENOMENA I, CHE 311

MIDTERM EXAM 3

CLOSED BOOK, TIME: 100 MINUTES

- 22/140
- Two solid spheres of identical material but different diameters R_1 and R_2 are allowed to sink freely in the same liquid. Calculate the ratio of their terminal velocities $\frac{v_2}{v_1}$, assuming $\frac{R_2}{R_1} = 4$ and that the friction factor for both is constant $f = 0.44$.

Hint: Drag force of the liquid of density ρ on the sphere of radius R and velocity v is $AfK = (\pi R^2) f \left(\frac{1}{2} \rho v^2 \right)$.

The buoyant force of the liquid on the sphere is $\frac{4}{3} \pi R^3 \rho g$.

- Consider a Newtonian fluid with constant viscosity μ and constant density ρ flowing between two large horizontal parallel plates, a distance $2b$ apart. For $t \leq 0$, flow is driven by a pressure gradient in z -direction according to the velocity profile: $v_z(x, t \leq 0) = v_{z,max} \left[1 - \left(\frac{x}{b} \right)^2 \right]$. At $t = 0$ the pressure gradient is turned off, after which the flow evolves in time towards stagnation.



- 22/60
- Simplify the equation of motion to derive a partial differential equation (PDE) for the velocity profile for $t > 0$.

$$\rho \left(\frac{\partial v_z}{\partial t} + v_x \frac{\partial v_z}{\partial x} + v_y \frac{\partial v_z}{\partial y} + v_z \frac{\partial v_z}{\partial z} \right) = - \frac{\partial p}{\partial z} + \mu \left[\frac{\partial^2 v_z}{\partial x^2} + \frac{\partial^2 v_z}{\partial y^2} + \frac{\partial^2 v_z}{\partial z^2} \right] + \rho g_z$$

- Rewrite the PDE in terms of dimensionless variables $\phi = \frac{v_z}{v_{z,max}}$, $\xi = \frac{x}{b}$, and $\tau = \frac{\mu}{\rho b^2} t$.

- Write down one initial condition and two boundary conditions for the dimensionless PDE.

- Solve the dimensionless PDE, knowing the solution may be written as an infinite sum of cosine functions $\phi(\xi, \tau) = \sum_{n=1}^{\infty} f_n(\tau) \cos\left(\frac{2n-1}{2} \pi \xi\right)$. Evaluate the integration constants using the orthogonality equation: $\int_{-1}^1 d\xi \cos(n\pi\xi) \cos(m\pi\xi) = \delta_{nm}$.

Hint: $\int_{-1}^{+1} \left[(1 - \xi^2) \cos\left(\frac{2n-1}{2} \pi \xi\right) \right] d\xi = -(-1)^n \frac{32}{[\pi(2n-1)]^3}$

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$$2.1 \quad \rho \frac{dv_z}{dt} = \mu \frac{\partial^2 v_z}{\partial x^2}$$

$$\tau = -\mu \frac{dv_z}{dt}$$

$$2.2 \quad \rho \frac{dv_z}{dt} = \frac{\mu}{\rho} \frac{\partial^2 v_z}{\partial x^2}$$

$$v_z = v_{z, \max} \left[1 - \left(\frac{x}{b} \right)^2 \right]$$

$$\phi = \frac{v_z}{v_{z, \max}} \quad \xi = \frac{x}{b} \quad \tau = \frac{\mu}{\rho b^2} t$$

$$v_z = v_{z, \max} t + C$$

$$v_z = v_{z, \max} [1 - \xi^2]$$

$$v_z \text{ at } x = 0$$

$$0 = v_z x + C$$

$$-v_{z, \max} \left[1 - \left(\frac{x}{b} \right)^2 \right] x = C$$

$$\frac{dv_z}{dt} = v_{z, \max} t - v_{z, \max} x$$

$$= v_{z, \max} \left(1 - \frac{x^2}{b^2} \right) - v_{z, \max} \left(1 - \frac{x^2}{b^2} \right)$$

$$1.1 \quad \frac{4}{3} \pi R^3 \rho g = \pi R^2 \rho \left(\frac{1}{2} v^2 \right)$$

$$\frac{8}{3} R g = -v^2$$

$$v_1 = \sqrt{\frac{8 R g}{3 f}}$$

$$v_2 = \sqrt{\frac{8 \cdot 4 \cdot R g}{3 f}} = \sqrt{\frac{32 R g}{3 f}}$$

$$v_2 = \frac{\sqrt{\frac{32 R g}{3 f}}}{\sqrt{\frac{8 R g}{3 f}}} = \sqrt{\frac{32 R g}{3 f} \cdot \frac{3 f}{8 R g}} = \sqrt{4} = 2$$

$$\boxed{\frac{v_2}{v_1} = 2}$$